Sonic Runner

CART263 — Project 2
Artist Statement 2

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Project Overview

Sonic Runner is a Sonic the Hedgehog-inspired 2D endless runner built using Phaser 3. The game features a single playable character, Sonic, in a Green Hill Zone-style environment. The player continuously moves forward as the world scrolls, and must survive by jumping over obstacles and pits, and using a spin attack to defeat enemies.

The game has no win condition. Instead, the goal is to survive for as long as possible while the speed gradually increases, making the experience more difficult over time.

Intentions

The goal of this project was to create a simple but complete game experience focused on fast, reaction-based gameplay. I wanted to explore how limiting the player's options to a small set of actions, such as jumping and using a spin attack, could still create a challenging and engaging system when combined with increasing speed and constant forward movement.

Another intention was to work within a controlled scope. Instead of trying to build a large or complex game, I focused on one character, one environment, and one enemy type. This allowed me to concentrate on refining the core mechanics and making sure the game felt responsive and consistent.

This project was also an opportunity to challenge myself by working outside of p5.js and using Phaser 3. I had some previous experience experimenting with game development in Godot, and certain aspects of Phaser, such as working with sprite sheets and animations, felt familiar. This made it a good choice for building a more structured game while still learning a new framework. At the same time, I wanted to explore an idea I had been interested in for a long time, which was creating a Sonic-inspired game.

Implementation

The game is built around a continuous auto-running system where the player remains in place while the world scrolls from right to left. The player can jump to avoid obstacles and pits, and use a spin attack to defeat enemies. If the player collides with an obstacle, falls into a pit, or touches an enemy without attacking, the game ends.

Enemies and hazards are spawned as the game progresses, and the speed of the game gradually increases over time. This creates a steady rise in difficulty and forces the player to react more quickly. A score system tracks how long the player survives, reinforcing the endless nature of the game.

Tools & Techniques

— Phaser 3 — game framework

— Arcade Physics — gravity, velocity, collision

— Sprite sheets and frame animations

— Adobe Photoshop — asset preparation and sprite cleanup

— Plain JavaScript — no additional frameworks

Key Challenges

Moving Physics Bodies
Phaser's static physics bodies are not designed to move, which caused issues when trying to scroll ground segments. This led to inconsistent collisions, so the solution was to use a single invisible ground line for physics while only moving the visual elements.

Sprite Sheet Slicing
The original Sonic Mania sprite sheet is not arranged on a consistent grid, making it difficult to slice automatically. To solve this, the necessary animation frames were manually extracted and reorganized into clean strips that Phaser could read properly.

Input Timing
Using Phaser's JustDown() input method only detects key presses for a single frame, which made it unreliable for a fast-paced game. Switching to a continuous input check with a cooldown made the spin attack more responsive and consistent.

Inspirations

Sonic the Hedgehog Series
The Sonic series was a direct inspiration for the character, visual style, and overall feel of movement. The focus on speed and the use of a spin attack influenced both the mechanics and the identity of the game.

Chrome Dinosaur Game
The Chrome Dinosaur Game inspired the core endless runner loop. The idea of continuous forward movement, simple controls, and increasing difficulty over time became the foundation for the gameplay structure.

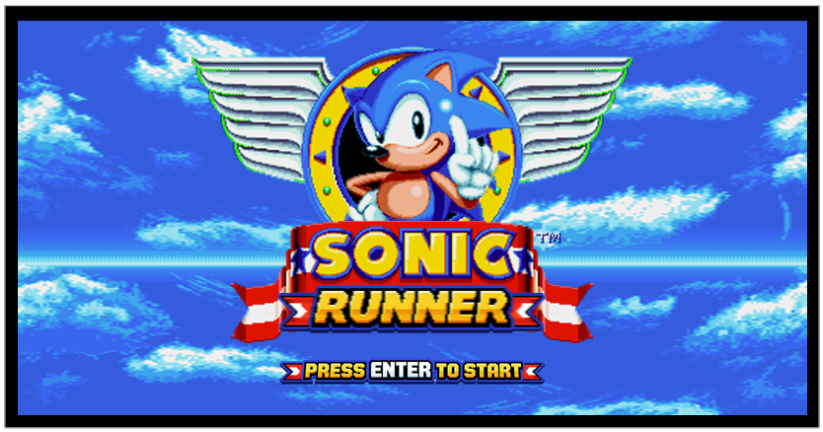
Frog Game (Previous Project)
My previous frog game project influenced my approach to scope and structure. It introduced me to working with game states, iteration, and building variations, which I applied here by developing the game in clear stages.

I Spy (Classmate aydanmcgrail Project)
This project influenced my approach to visuals and interaction. It showed how simple and recognizable elements could still create an engaging experience, which pushed me to keep the visual design straightforward while focusing on gameplay.

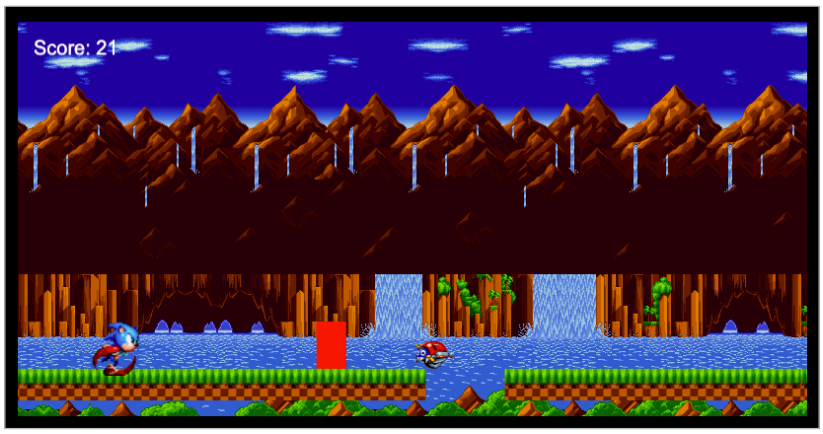
Attributions

Sonic character sprites were sourced from Sonic Mania sprite sheets (ripped by AsuharaMoon).
Motobug enemy sprites were sourced from Sonic Mania assets (arranged by Modern, sprites by Dolphman).
Green Hill Zone background and tile assets were used for the environment.
Sound effects (jump, spin, enemy defeat, game over) and music tracks were sourced from Youtube and originate from Sonic Mania. Credits to SEGA

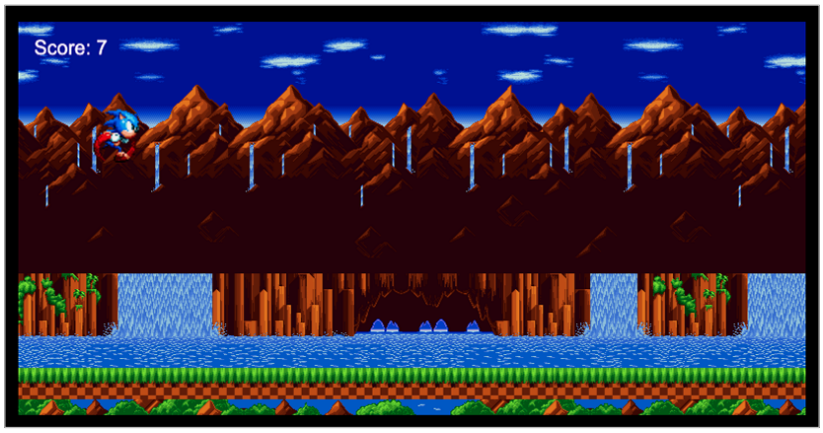
Screenshots



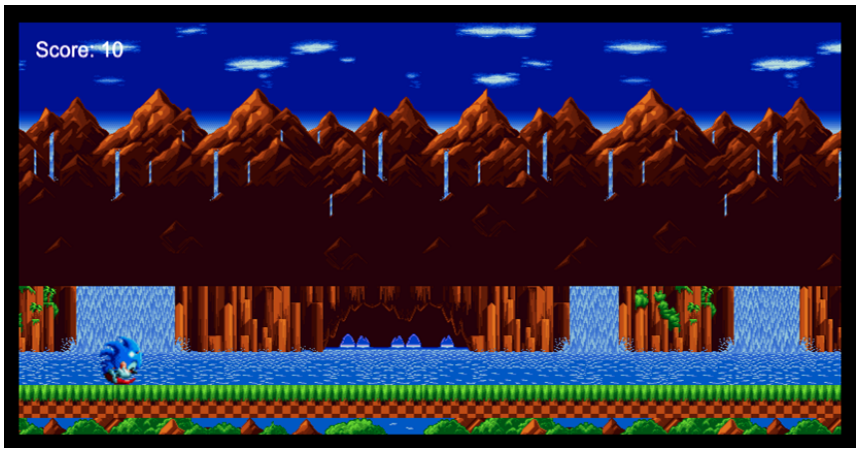
Title Screen — the opening screen of the game.



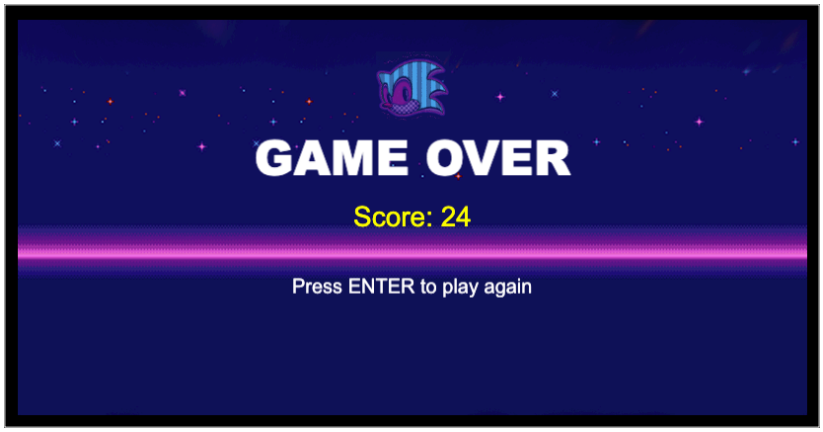
Gameplay Overview — Sonic running through Green Hill Zone.



Jumping — player jumping over an obstacle



Spin Attack — Sonic spinning to defeat a Motobug.



Game Over Screen — final score displayed after dying.